

MD HASAN ANOWAR

Ames, IA 50010

hasan92niloy@gmail.com | 956-360-1171 | [linkedin.com/in/hasananowar](https://www.linkedin.com/in/hasananowar) | github.com/hasananowar

PROFESSIONAL SUMMARY

4th year PhD Student in computer engineering with 5 years of expertise in machine & deep learning, data mining, and information retrieval. Performed applications of neural networks, generative AI (Diffusion, LLM), statistical modeling, and time series analysis. Also proficient in databases, distributed systems, and large-scale data analysis tools. Documented 2 peer-reviewed research papers as first author.

EXPERIENCES

Machine Learning Engineer Intern (incoming) | Summer 2024 | ViaSat, CA

Deep Learning Researcher | Spring 2023-Spring 2024 | Iowa State University, IA

Graph Representation Learning

- Engineered dynamic graph structures via data augmentation of time-stamped spatial datasets.
- Developed transformer encoder models for evolving graphs using PyTorch and PyTorch-Geometric.
- Successfully predicted dynamic graph edges, demonstrating proficiency on benchmark recommender datasets compared to static and dynamic GNN baselines, e.g., GAT and HTGNN.

Sentiment Analysis (RNN)

- Designed a sentiment detection model for movie reviews using the sequence models from PyTorch.
- LSTM based model elevated the accuracy by outperforming vanilla RNN by 50%.

Image Style Transfer (CNN)

- Tasked to transfer style from an art image to a content image. Implemented convolutional neural network model (VGG19) from TensorFlow and Keras to extract the feature map.
- Introduced an additional convolution layer and fine-tuned feature weights resulting in obtaining the visually optimal image.

Machine Learning Researcher | Jan 2019-Dec 2020 | University of Texas Rio Grande Valley, TX

- Developed a Python-based software tool for brain-computer interface to interpret biological signals.
- Performed predictive modeling incorporating supervised learning models such as support vector machine, decision trees, logistic regression, linear discriminant analysis, KNN, etc.

Data Science Researcher | Spring 2021-Present | Iowa State University, IA

- Responsible for uncovering spatial-temporal patterns from large moving object data (>2,000 GBs).
- Conducted cleaning and processing of raw chemical data utilizing Pandas, SciPy, and Numpy.
- Applied unsupervised learning techniques, employing clustering algorithms (DBSCAN, Hierarchical, K-Means) for atomic motion analysis, leading to a novel spatiotemporal cluster.
- Developed a compression algorithm to store huge volumes of trajectory data, enhancing query processing efficiency by 4 times and storage gain by 5 times.
- Collaborated with chemists on capturing causal relationships based on domain semantics.

Large-Scale Data Systems Instructor | Fall 2023 - Present | Iowa State University, IA

- Taught lab sessions on static and dynamic large-scale data processing and related algorithm designs.
- Integrated Java with big data tools like Hadoop's MapReduce for static data and Apache Flink and Apache Spark for dynamic data.
- Conducted hands-on experiences with Apache Pig, RDDs, and HDFS.

EDUCATION

Ph.D. in Computer Engineering | Jan 2021-May 2025 (expected)

Iowa State University (ISU), IA, USA

M.S. in Electrical Engineering | Jan 2019-Dec 2020

University of Texas Rio Grande Valley (UTRGV), TX, USA

B.Sc. in Electrical and Electrical Engineering | May 2010-Sept 2015

Bangladesh University of Engineering and Technology (BUET), Bangladesh

PROJECTS

Integration of Chameleon Cloud with LLVM Compilation (Cloud Computing)

- Provisioned a bare-metal node and launched a bare-metal instance.
- Built a functional docker container from image, compiled a C program into intermediate representation (IR) code using LLVM/Clang, and deployed it in the cloud.

Database and Query Processing | Fall 2022 | Iowa State University, IA

- Performed Transaction SQL queries using MySQL Workbench.
- Designed multidimensional cubes using SQL Server Data Tools and SQL Server Management Studio and performed OLAP using MDX on large volume of data from the data warehouse.
- Worked with NoSQL database management system (Neo4j, MongoDB) and geospatial data using geographic system application (QGIS).

Senior Design Project Mentor | Fall 2023 - Spring 2024 | Iowa State University, IA

- Mentored six undergrad students to develop a user interface for visualization of mined patterns from trajectory data.

SKILLS

Programming: Python, Java, C/C++, MATLAB

Python Libraries: PyTorch, Keras, TensorFlow, Scikit-learn, Pandas, Numpy, SciPy, DGL, PyTorch Geometric, NetworkX

Database and Big Data: SQL, NoSQL, MySQL Workbench, PostgreSQL, MongoDB, Apache Spark, Hadoop, MapReduce, Pig, QGIS, MDX, Cypher Neo4j, Flink

Data Visualization: Matplotlib, Seaborn, Plotly, PowerBI, Tableau, Excel

Platform: Linux, MacOS, Windows, VSCode, PyCharm

Miscellaneous: Git, AWS SageMaker, Google Colab, Jupyter Notebook, CUDA, Docker, LaTeX

PUBLICATIONS

- Anowar, M.H. et al. Generalization Aware Compression of Molecular Trajectories. ADBIS 2022.

LNCS, vol 13389. Springer, Cham. **(Best Paper Award)**

- Anowar, M.H., Roy, P., "A Modified Incremental Conductance Based Photovoltaic MPPT Charge Controller" in ECCE'19

LEADERSHIP SKILLS

Vice-President, GO-ECpE at Iowa State University | Fall 2022-Present

- Monitoring the club's financial activities and writing funding proposals.
- Organizing technical and social activities for professional development, e.g., LaTeX workshop.